Distributed Human Activity Recognition

Kingshuk Goswami

Computer Science and Engineering
Dr. B. C. Roy Engineering College, Durgapur

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Distributed Human Activity Recognition

Human Activity Recognition

Human Activity Recognition refers to the process of using sensor data from various devices such as smartphones and wearable devices to detect and categorize human activities such as walking, running, cooking, etc.

However, these sensor data may contain sensitive information about an individual's behavior, routines, location, health status, and lifestyle.

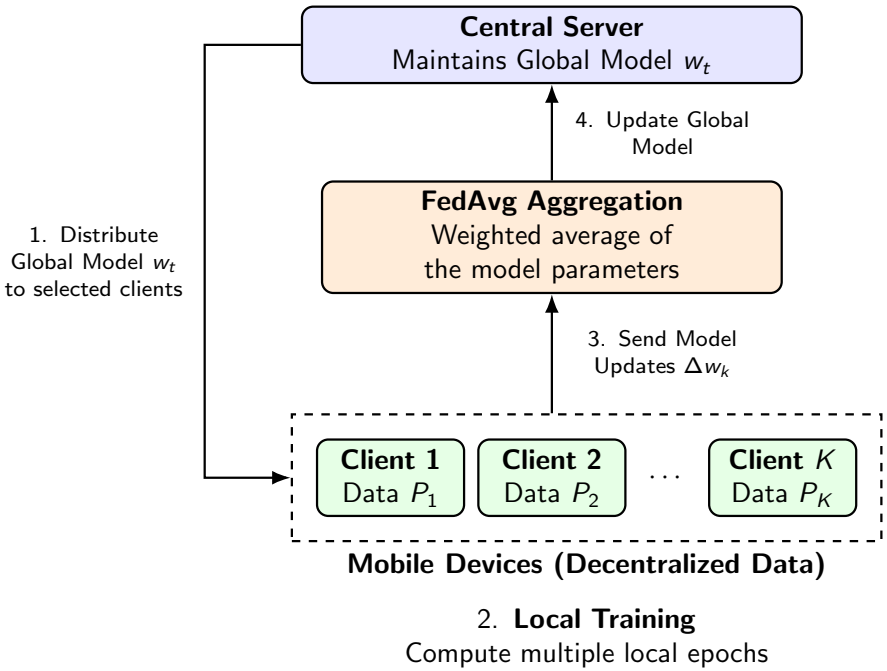
A straightforward solution is **centralized learning**, in which data collected from different users are transferred to a central server and jointly used to train a model. However, this approach is impractical when faced with **limited time or computing resources**, or if access to data is restricted due to **privacy concerns**.

Research Problem

How can a model be collaboratively trained across multiple users without transferring their raw sensor data to a central server?

Federated Learning(FedAvg)

FedAvg (Federated Averaging) is a foundational FL algorithm in which the server sends the global model to selected clients, and each client performs local training for a fixed number of epochs. The local parameters are weighted-averaged at the server. A client with more local training data has greater importance in the global objective.



It faces challenges in **heterogeneous federated environments**, *where clients may have different data distributions. Thus, each client optimizes a different local objective. With many local updates, their models can move in different directions, leading to* **client drift**.

FedProx: Addressing Heterogeneity

1. Statistical Heterogeneity

Clients have different data distributions. In HAR, one client's data may be dominated by *running*, while another client's data may be dominated by *sitting*.

FedProx introduces a **proximal term** into the local objective, which **restricts local model updates from moving too far from the global model** and thereby reduces **client drift**.

2. Systems Heterogeneity

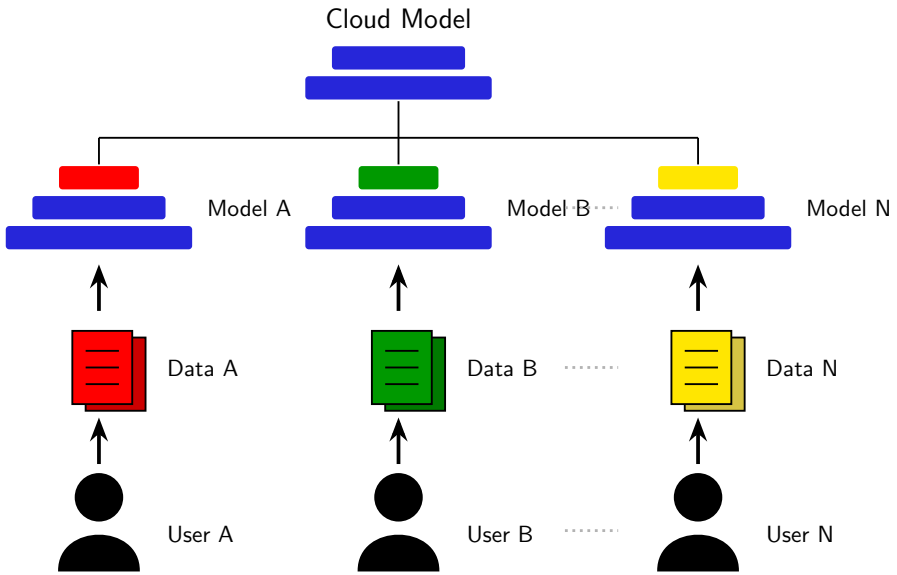
The devices of the clients are different. They differ in computation capability, battery availability, communication speed, etc.

FedProx introduces **tolerating partial work**, allowing clients to perform different amounts of local computation and incorporating **partial solutions** from stragglers instead of simply discarding them.

Overall, FedProx provides better stability under heterogeneous clients, but it still learns a **single global** *model and therefore does not provide explicit client-level personalisation.*

FedPer: Personalization Layers

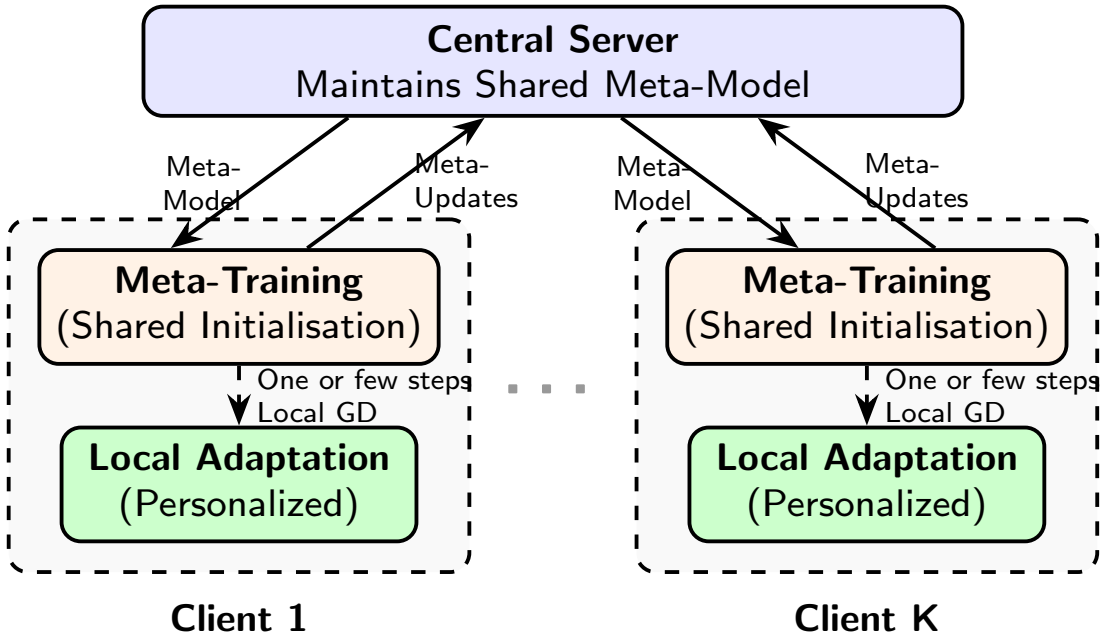
FedPer addresses statistical heterogeneity in federated learning for **deep neural networks**. It divides the network into **shared base layers** and **client-specific personalization layers**. The base and personalization layers are trained locally on each client. After local training, only the shared base layers are uploaded and aggregated using federated averaging, while the personalization layers remain private to each client. Thus, the base layers capture **common knowledge**, while personalization layers capture **user-specific characteristics**.



Its performance depends heavily on **choosing the right division between shared and personalisation layers** *and is* **limited to neural-network architectures**.

Per-FedAvg: Meta-Learning Personalization

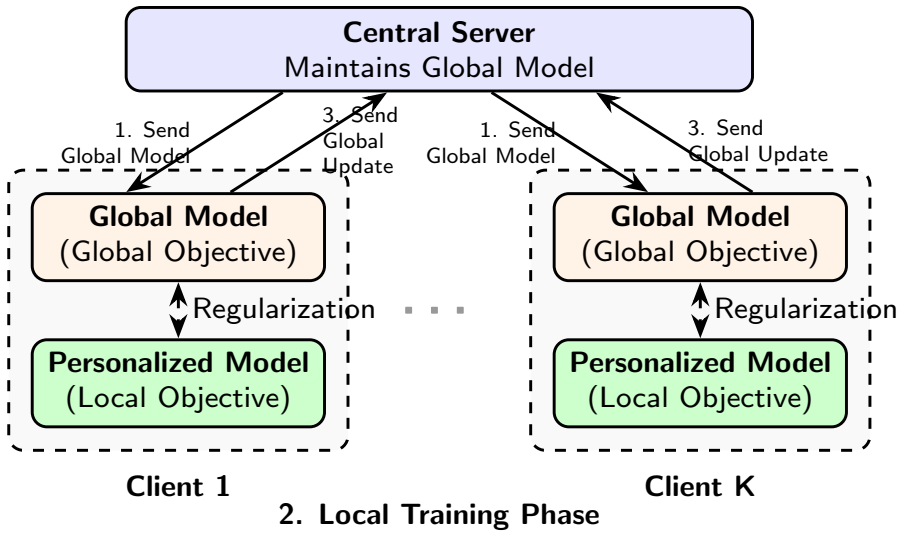
The paper proposed **Per-FedAvg**, which formulates personalized federated learning using the **MAML framework**. Instead of a directly used global model, it learns a shared **meta-model** that can be efficiently adapted to each client. Each client performs one or a few gradient-descent steps from this shared initialization to obtain its personalized model.



The main advantage is that personalization is achieved through **local adaptation without dividing the network into shared and private layers**. *However, its* **meta-learning formulation adds optimization complexity**, *making it computationally expensive for large neural networks.*

Ditto: Federated Multi-Task Personalization

Ditto proposed a **federated multi-task learning framework** that maintains both a global model and a personalised model for each client. Each personalised model is trained on local data while being **regularized toward the global model**, allowing clients to adapt to heterogeneous data while benefiting from shared knowledge. Ditto further addresses **accuracy, fairness, and robustness** and serves as a **lightweight, modular personalisation layer** that can be added to standard federated learning.



2. Local Training Phase

Clients alternately solve the global objective and a regularized local objective

Its global-and-local structure enables personalisation while retaining the standard federated learning framework. However, personalised models add storage and computation, while regularization strength must balance personalisation and global knowledge.

Selecting the Personalised FL Architecture

Deployment Constraint

HAR on resource-constrained edge devices: Flash < 8 MB, limited RAM, computation and communication capability, with heterogeneous client data.

Best Fit

FedPer: it provides client-level personalisation while keeping only a small set of personalisation layers private, making it suitable for resource-constrained devices.
FedAvg: No explicit client-level personalisation but use as baseline.

Per-FedAvg: Provides personalisation through local adaptation without requiring separate shared and private layers.

Why Not the Others?

FedProx: Handles heterogeneity but does not provide personalisation.

Ditto: Maintains an additional personalised model, increasing local storage and computation.

Privacy Challenges in Federated Learning

The Privacy Problem

Federated Learning keeps **raw sensor data on the client**, but **model updates are shared with the server**. Since these updates are computed from private local data, they may still leak information about individual clients.

What can be leaked?

- **Membership / Property Inference:** An attacker may infer whether a particular record or property is present in a client's training data.
- **Data Reconstruction:** Inputs and labels can potentially be reconstructed directly from shared gradients.

Privacy-Preserving Techniques

- **Secure Aggregation**
- **Differential Privacy**

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Thank You